

Progress Report Sprint 10

- Course Name: CS461 Senior Software Engineering Project I
- Date: 03/12/26
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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Summary - Outcomes, blockers, and insights

- **UI:** Made informed changes to the UI based off of second-phase interviewing, updated styling and made CSS code more consistent; updated the javascript to call the look-up table and regressor microservices instead of the prototype microservice.
- Updated model architecture to be able to handle unknown taxonomy fields.
- Created lookup table microservice which references NCBI and GBIF databases to gather taxonomy data.
- Created test scripts for the APIs for lookup table microservice and regression model microservice.

Blockers:

- A big blocker we discovered while making our demo was that render's free tier does not allow for enough memory usage to support our regressor microservice.

Work Log

1. Second-Phase Interviewing and Subsequent UI Cosmetic Updates

Status: In-progress

Owner: Hailey

Description: Did a second round of interviewing because I really enjoyed the impact that the last round had on the UI. Got more feedback on the cosmetics rather than the function this time, which I appreciated. Implemented the recommendation to lessen whitespace, but otherwise will continue implementing the feedback next term.

Source Link: [w 2026-02-25-CS.032-Sprint9-ProgressReport.docx](#)

Supporting Artifact: Item.01 and Item.02

Notes: This round of interviewing took the back burner because of the JS updates and demo, so not all suggestions have been addressed yet.

2. Updated API Call in 'index.js' and New Render Web Services

Status: In-progress

Owner: Hailey

Description: Created two new web services on render to handle the new look-up table and regressor microservices. Altered the javascript code to interact with the look-up-table and regression-model web services instead of the prototype. Deployed the look-up-table successfully, but encountered the memory problem with render's free tier and the regression-model. I temporarily changed the javascript to simply display the taxonomic information returned from the look-up table rather than sending it to the regressor service.

Source Link: [2026-02-12-CS.032-Sprint8-ProgressReport.docx](#)

Supporting Artifact: Item.03

3. Regression Microservice

Status: In-progress

Owner: Grant

Description: The regressor microservice uses a static, trained version of the XGBoost model to predict an input species's mass based on an input taxonomy. The route for single species mass-prediction is complete. However, more work is needed to fully implement a route for a list of predictions.

Source Link: [2026-01-13-CS.032-Sprint6-ProgressReport.docx](#)

Supporting Artifacts: Item.04, Item.05

Notes: The code is fully implemented and merged into the main branch of the GitHub repository. However, the current model is 1.8 GB, so more server hardware is needed before deployment.

4. Lookup Table Microservice

Status: In-progress

Owner: Grant

Description: The lookup table microservice uses NCBI and GBIF databases to retrieve input species taxonomies using the appropriate fuzzymatcher API. The route for single species taxonomy gathering is functional. However, more work is needed to incorporate taxonomy gathering for multiple species. This microservice has been successfully integrated into the existing deployment.

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts: Item.04, Item.05

Notes: For information about the API, check out the manpages here: [NCBI API](#), [GBIF API](#)

5. API Testing and Calling

Status: In-progress

Owner: Anish

Description: The code for the API is complete, we just to pass some instructions into the API and see if the json objects are being generated from the data set. Also need to setup with the variable in Xbooster so we can run them parallely and give out both outputs.

Source Link: [2026-01-30-CS.032-Sprint7-ProgressReport.docx](#)

Supporting Artifacts:

Notes: For information about the API, check out the manpages here: [NCBI API](#)

Risk and Quality Tracking

Open Bug Count: 4

TOP BUGS	SEVERITY	OWNER(S)	STATUS	LINK	ACTIONS
DATASET TOO LARGE FOR DECISION TREE	High	Grant	Completed	Decision Tree Code on GitHub	The model was successfully able to run with the dataset of about 37,000 species without issue. However, this issue extends into the bug titled, "Regressor Microservice Too Big For Render Free Tier."
NON-RESPONSIVE CSS	Low	Hailey	In-progress	primary CSS file on GitHub	Hailey will add in CSS that better accommodates smaller screen sizes (media queries) by the end of this term (not a top priority)
CACHED QUESTIONS ARE NOT LOADING	Low	Hailey	In-progress	'More questions' service python file on GitHub	Hailey will debug the API call and JS code and then, if necessary, see if it is an issue with the render deployment
REGRESSOR MICROSERVICE TOO BIG FOR RENDER FREE TIER	High	Grant and Hailey	In-progress	Render deployment status	We will contact the project partner about accessing dedicated hardware with enough on board memory to handle the Regressor Microservice.

Next Sprint Plan

- We need to problem solve our microservice deployment and get body mass predictions displayed to the front-end. This could mean simplifying the regressor microservice or accessing a more powerful deployment option. This, again, works toward pretty much every functionality-based requirement from the beginning of last term.

Priority: High; Link: [2025-10-23-CS.032-Requirements](#)

- We will begin work on incorporating multi-species predictions into both the lookup table and the regressor microservice. This is in line with REQ-003 which indicates that the user must be able to make at least 10 simultaneous queries.

Priority: Med; Link: [2025-10-23-CS.032-Requirements](#) (REQ-003)

- We will also begin researching how to incorporate confidence intervals into our predictions. This is to fulfill REQ-001 which requires some sort of confidence interval with each prediction.

Priority: High; Link: [2025-10-23-CS.032-Requirements](#) (REQ-001)

Team Reflection

This sprint was a big step for our group. We actually saw the implementation of a different microservice than the prototype lookup; although the implementation is not complete because of a bug, this should be a quick fix at the beginning of the next term. We are excited to show our project partner the return of taxonomic information as a result of the look-up table microservice.

Naturally, fixing the bug in the regressor microservice is our primary goal for the next sprint, it will take a little bit of problem solving to determine which route we want to take to fix the bug (either minimizing the microservice or increasing the capabilities of our deployment service). We will know that our goal is accomplished when the regressor microservice deploys successfully on render.

As a group, it was exciting to have the opportunity to present our project and progress in lecture. This gave us the chance to recognize exactly where each of our strengths and weaknesses were as they related to the motivations and functionality behind the project. We can all stand to learn more about what each other are working on, even if it is not necessarily our specialty.

Artifacts Appendix

Item.01

[TaxonBodyMass Third-Party UI Interview](#)

This is the document I have used to track questions and recommendations during both phases of interviewing; some things on this document have not been implemented yet, hence why I considered this phase of interviewing “in-progress”.

Item.02

[Video Depicting Old Styling](#); [Video of Updated Styling](#)

(Left): Video from past sprint of the tutorial styling, included here to show how the modal styling used to look. (Right): Video of new modal styling done to adhere to the interviewing recommendation of “lessening white space”.

Item.03

[Video of Look-Up Table Microservice Response](#)

This video depicts the updated response after inputting a species name; this is currently the taxonomic information of the species because of the open bug concerning the regressor microservice. When that bug gets patched, the taxonomic information displayed will instead be sent to the regressor microservice, which will itself return the predicted body mass.

Item.04

```
{
  "type": "object",
  "taxonomy": {
    "kingdom": { "type": "string" },
    "phylum": { "type": "string" },
    "class": { "type": "string" },
    "order": { "type": "string" },
    "family": { "type": "string" },
    "genus": { "type": "string" },
    "species": { "type": "string" },
  }
}
```

Fig.01 - This is the schema for the input to /xgb_pred_single, the single species prediction route for the regressor microservice. It is also the output for /single_species taxonomy gathering route for the lookup table microservice.

Item.05

```
{
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "taxonomy": {
        "type": "object",
        "properties": {
          "kingdom": { "type": "string" },
          "phylum": { "type": "string" },
          "class": { "type": "string" },
          "order": { "type": "string" },
          "family": { "type": "string" },
          "genus": { "type": "string" },
          "species": { "type": "string" }
        }
      }
    }
  }
}
```

Fig.02 - This is the schema for the input to /xgb_pred_multi, the single species prediction route for the regressor microservice. It is also the output for /multi_species taxonomy gathering route for the lookup table microservice.